

CO3 AT1-Case Study

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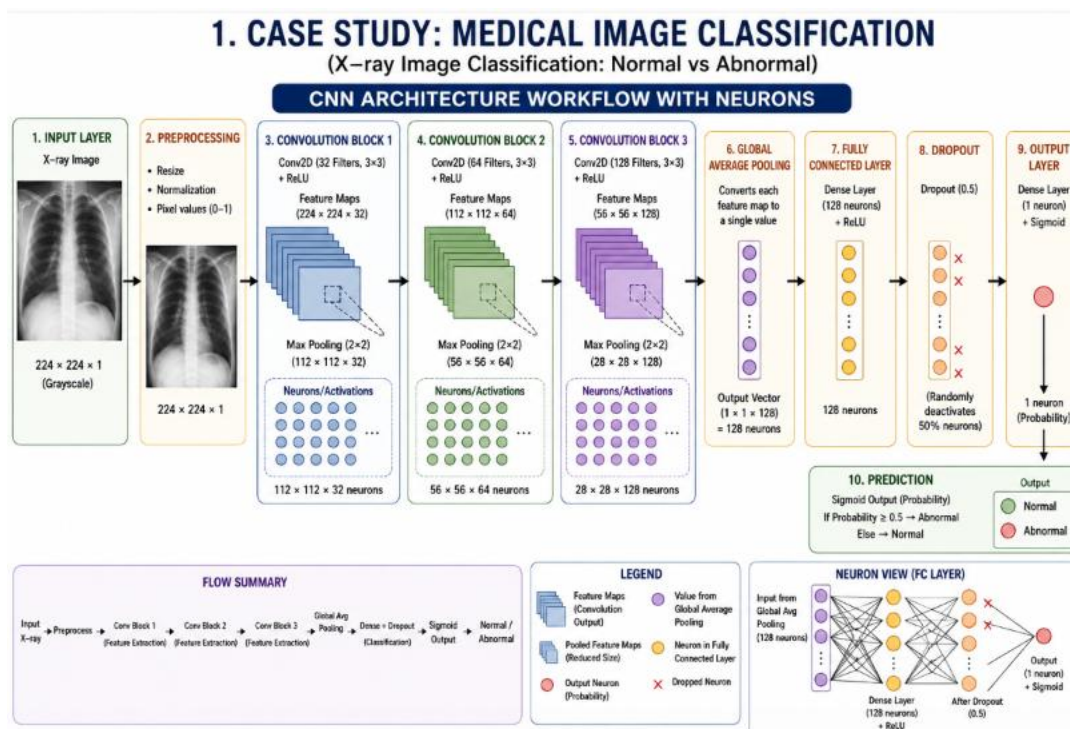
Question 1: Medical Image Classification

Explanation

Medical image classification uses a Convolutional Neural Network (CNN) to classify X-ray images into categories such as Normal and Abnormal. The input X-ray is first resized and normalized. Convolution layers extract important features such as edges, textures and abnormal regions, while pooling layers reduce the size of the feature maps.

The extracted features are passed through fully connected layers, and a Sigmoid activation function produces the final classification. The system can help doctors analyze large numbers of X-ray images efficiently, but the model should be properly validated on real clinical datasets before medical use.

ARCHITECTURE DIAGRAM



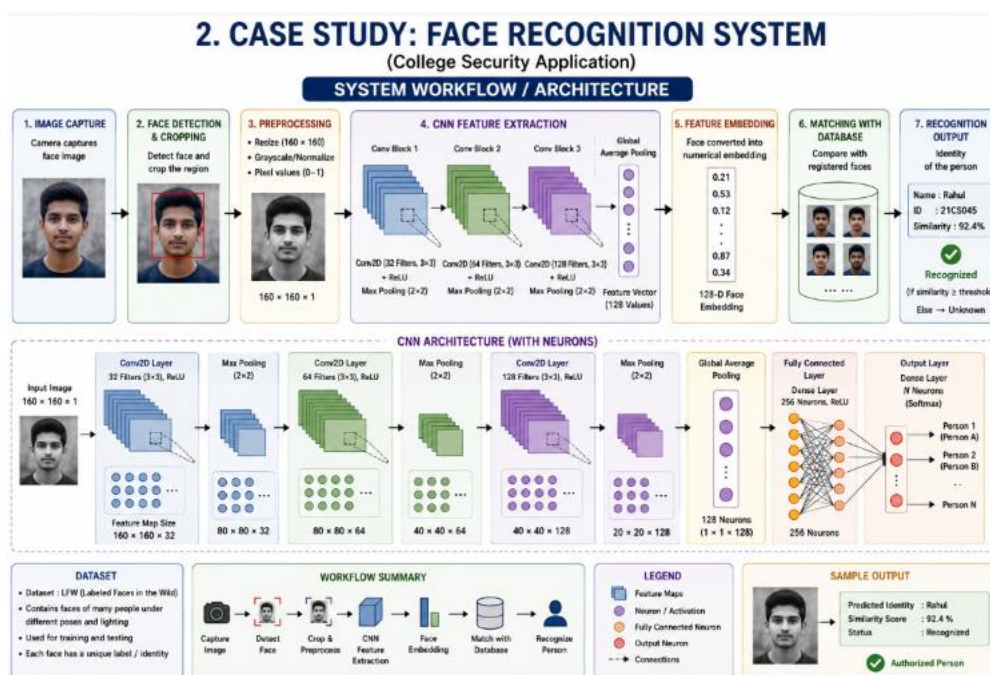
2. Face Recognition System

Explanation

A CNN-based face recognition system is used to identify or verify a person from a facial image. The system first detects and preprocesses the face, then convolution layers automatically extract important facial features such as the eyes, nose, mouth and overall face structure.

The extracted features are converted into a representation called a face embedding, which can be compared with registered face representations. The final system identifies the closest matching person or returns an unknown result. In real-world applications, privacy, consent and secure storage of facial data are very important.

ARCHITECTURE DIAGRAM

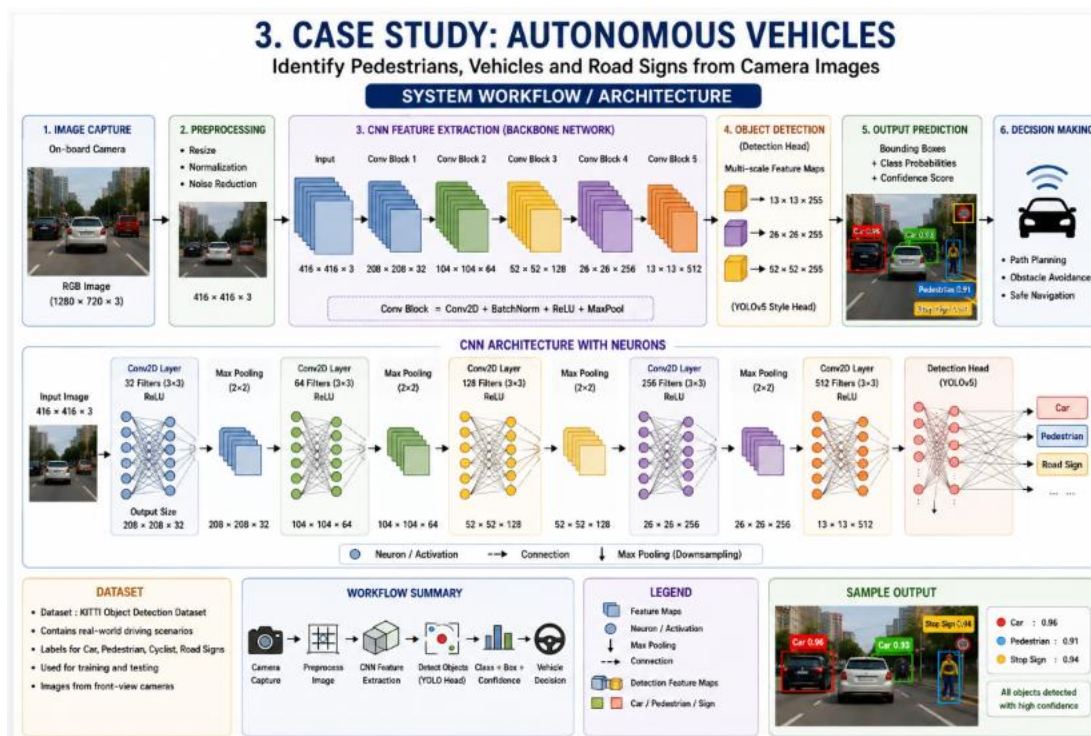


3. Autonomous Vehicle Using CNN

Autonomous vehicles use computer vision and CNNs to understand objects present on the road. Images captured by cameras are preprocessed and passed through convolution layers, which extract features such as shapes, edges and textures. The system can classify objects such as cars, pedestrians and stop signs.

The detected object and its confidence score are then used by the vehicle's decision-making system. CNN-based vision helps the vehicle understand its surroundings and supports safer navigation. For a real autonomous vehicle, more advanced object-detection models are generally required to locate multiple objects using bounding boxes.

ARCHITECTURE DIAGRAM

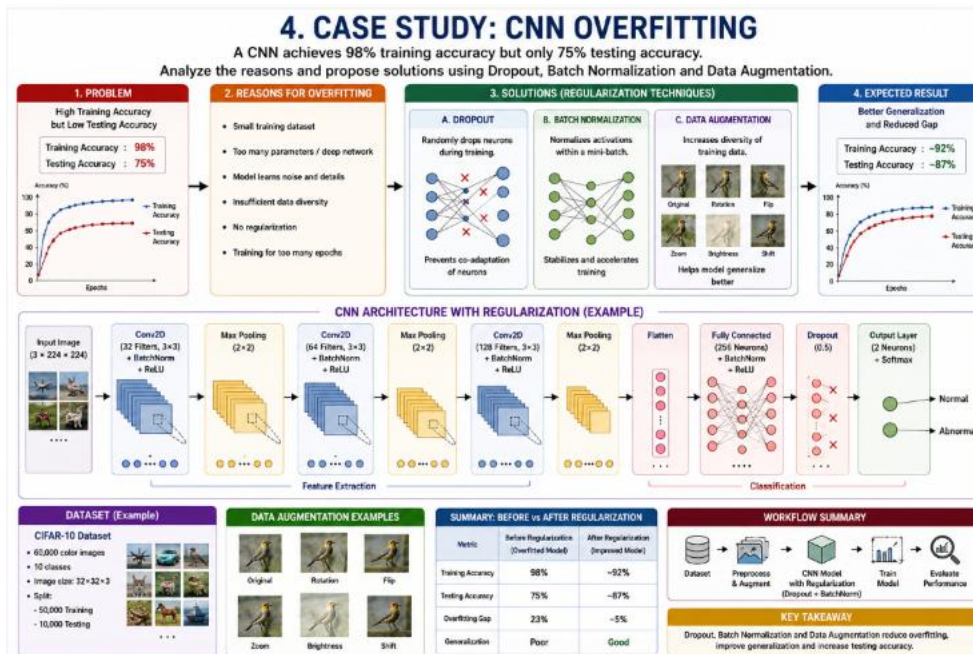


4. CNN Overfitting and Its Reduction

Overfitting occurs when a CNN performs very well on training data but performs poorly on unseen testing data. For example, a model may achieve 98% training accuracy but only 75% testing accuracy. This means the model has learned the training examples too closely instead of learning general patterns.

Overfitting can be reduced using techniques such as Dropout, Batch Normalization, Data Augmentation and Early Stopping. These methods help the model generalize better to new images. Training and validation accuracy/loss graphs are useful for identifying and analyzing overfitting.

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Question 5 . Fruit Image Classification Using CNN

Fruit image classification uses a CNN to identify different fruits from images. In this case, the model can classify Apple, Banana, Orange and Mango. The input image is resized and normalized, and convolution layers extract visual features such as color, shape, edges and texture.

Pooling layers reduce the feature-map size, while dense layers use the extracted features for classification. A Softmax activation function produces the probability of each fruit class, and the class with the highest probability becomes the predicted fruit. CNNs can therefore automate fruit identification and sorting

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